Electrostatics and Electric flux

Fri May 08 2026

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Electrostatics

Grade 12 • Physics • Session 2





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Recap

- Introduction to Electrostatics
- Electric Charge
- Properties of Electric Charge
  - Additivity
  - Conservation
  - Quantisation
- Conductors and Insulators
- Methods of charging a body
  - Friction (rubbing)
  - Conduction
  - Induction
- Coulomb's Law

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Force due to multiple Charges

F_{Net} on 2C = ?

$F_1(2C \& 4C) = \frac{k(4)(2)}{a^2} = \frac{8k}{a^2}$

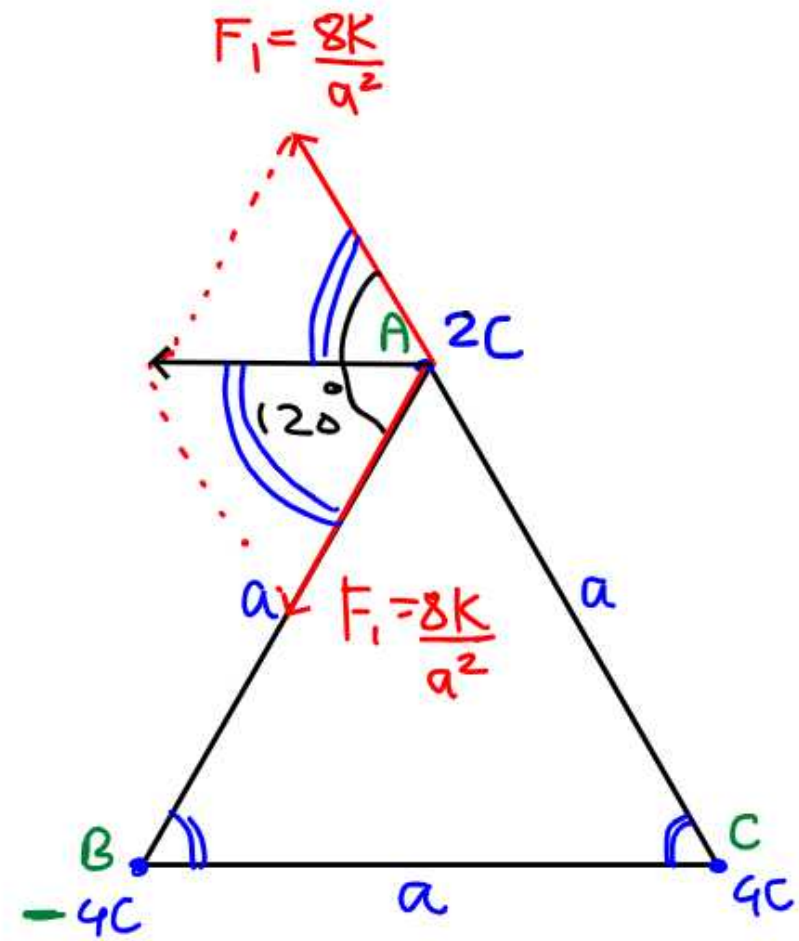
$F_2(2C \& 4C) = \frac{k(4)(2)}{a^2} = \frac{8k}{a^2}$

(A) $8k/a^2$

(B) $16k/a^2$

(C) $4k/a^2$

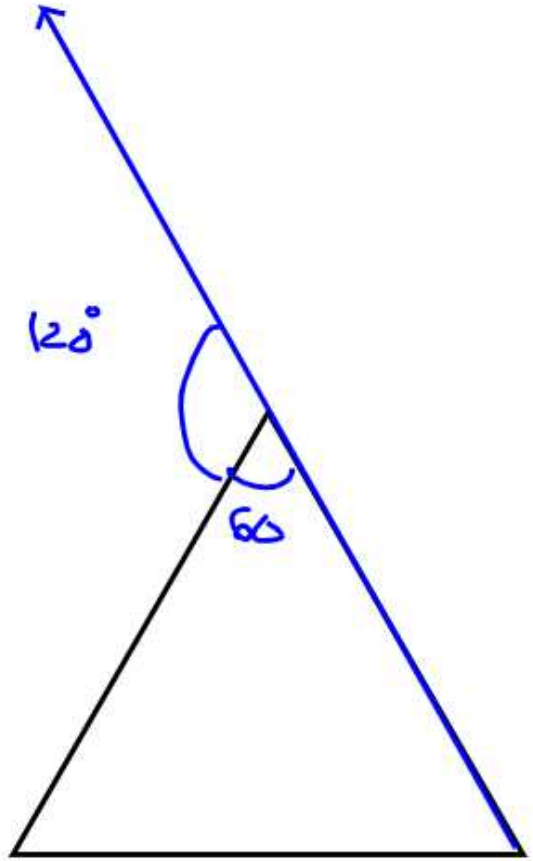
(D) $4\sqrt{2}k/a^2$



$F_{Net} = 2F_1 \cos\left(\frac{\theta}{2}\right)$

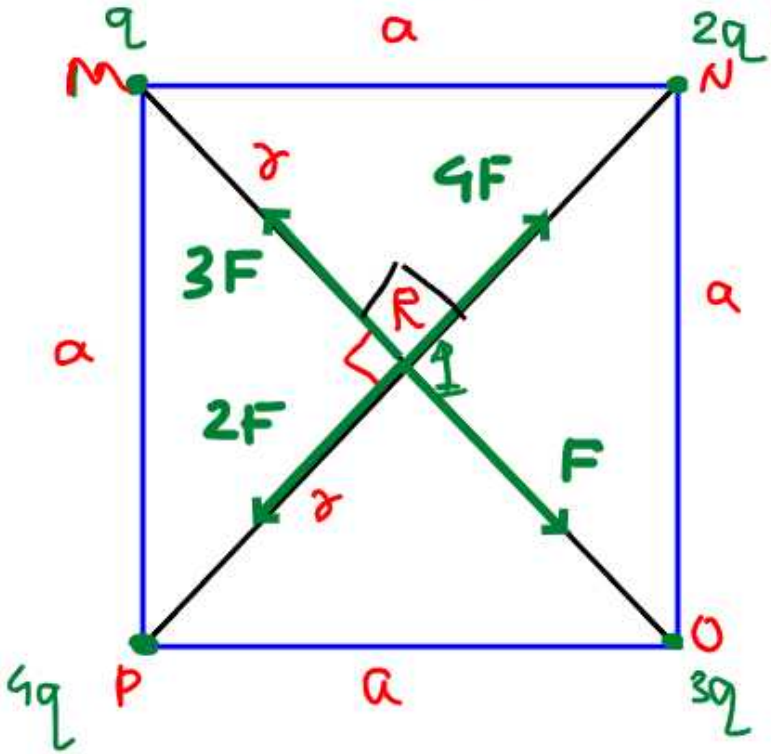
$= 2F_1 \cos\left(\frac{120^\circ}{2}\right)$

$= F_1 = \frac{8K}{a^2}$


120°
60

$r^2 + r^2 = a^2 \rightarrow 2r^2 = a^2$

$r = \frac{a}{\sqrt{2}}$



Find net force on 1 unit charge.

(A) $\frac{4\sqrt{2}kq}{3a^2}$

(B) $\frac{2\sqrt{2}kq}{a^2}$

(C) $\frac{4kq}{a^2}$

(D) $\frac{4\sqrt{2}kq}{a^2}$ ✓

$F_{Net} = \sqrt{(2F)^2 + (2F)^2}$

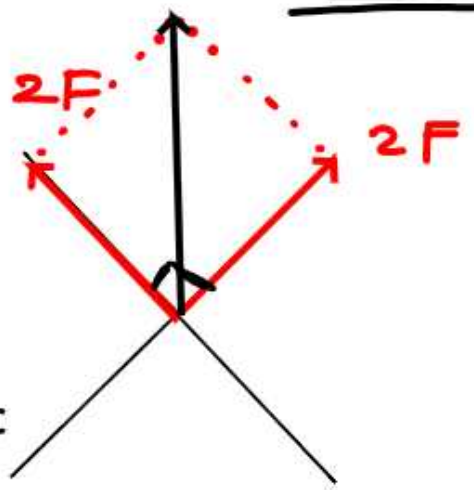
$= \sqrt{2(2F)^2}$

$= 2F\sqrt{2}$

$F_{1q} = \frac{kq(1)}{r^2} = \frac{kq}{\left(\frac{a}{\sqrt{2}}\right)^2} = \frac{2kq}{a^2}$

$F_{Net} = \sqrt{2}(2F) = 2\sqrt{2}\left(\frac{2kq}{a^2}\right)$

$= \frac{4\sqrt{2}kq}{a^2}$



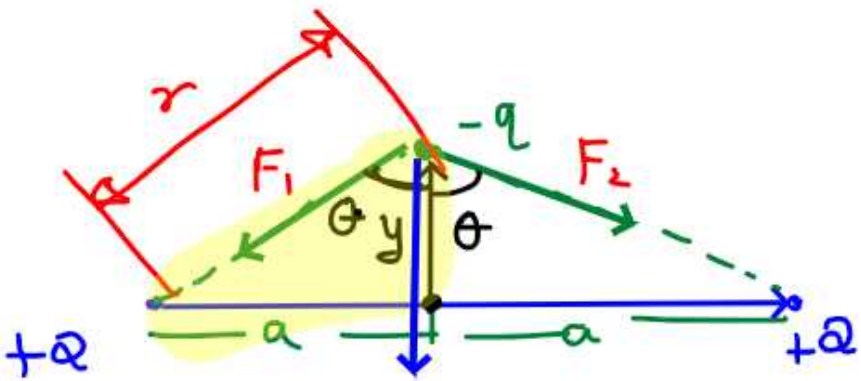
$F_{Net} = 2(2F)\cos\left(\frac{90}{2}\right)$

$= 2(2F)\frac{1}{\sqrt{2}}\frac{\sqrt{2}}{\sqrt{2}} = 2\sqrt{2}F$

$r^2 = y^2 + a^2 \Rightarrow r = (y^2+a^2)^{1/2}$

$y \ll a$

$F \propto -y$





$F \propto y$

✓ $F \propto -y$

$F \propto \frac{1}{y}$

$F \propto -\frac{1}{y}$

$F_1 = \frac{k(Q)(q)}{r^2}$

$F_1 = \frac{kQq}{y^2+a^2} = F_2$

$F_{Net} = 2F_1\cos\left(\frac{\theta}{2}\right)$

$= 2F_1\cos\theta$

$= 2\left(\frac{kQq}{y^2+a^2}\right)\left[\frac{y}{(y^2+a^2)^{1/2}}\right]$

$F_{Net} = \frac{2kQqy}{(y^2+a^2)^{3/2}}$

$y \ll a, \therefore y^2 + a^2 \approx a^2$

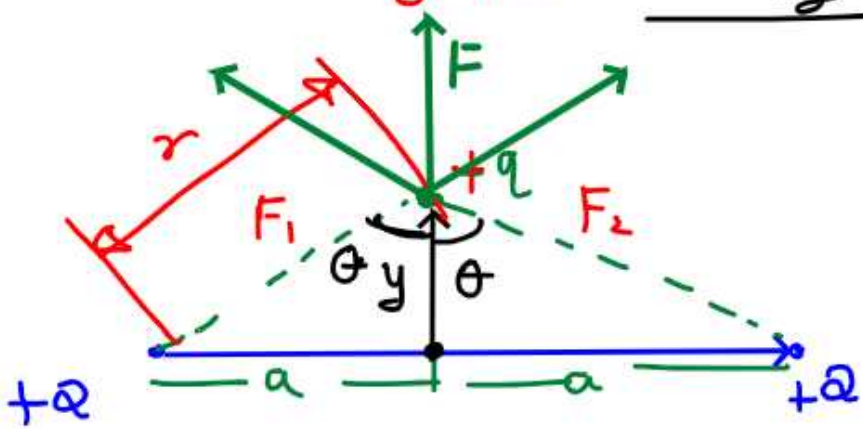
$F_{Net} = \frac{2kQqy}{(a^2)^{3/2}}$

$\vec{F}_{Net} = -\left(\frac{2kQq}{a^3}\right)\vec{y}$

$r^2 = y^2 + a^2 \Rightarrow r = (y^2+a^2)^{1/2}$

$y \ll a$

$F \propto -y$



✓ $F \propto y$

$F \propto -y$

$F \propto \frac{1}{y}$

$F \propto -\frac{1}{y}$

F_{Net}

$F_1 = \frac{k(Q)(q)}{r^2}$

$F_1 = \frac{kQq}{y^2+a^2} = F_2$

$F_{Net} = 2F_1\cos\left(\frac{\theta}{2}\right)$

$= 2F_1\cos\theta$

$= 2\left(\frac{kQq}{y^2+a^2}\right)\left[\frac{y}{(y^2+a^2)^{1/2}}\right]$

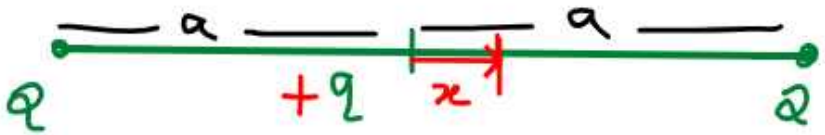
$F_{Net} = \frac{2kQqy}{(y^2+a^2)^{3/2}}$

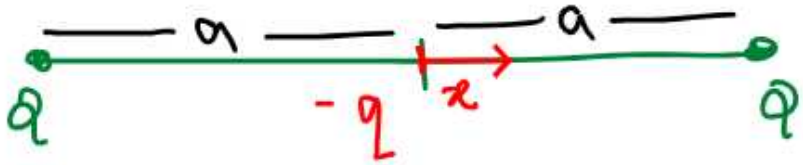
$y \ll a, \therefore y^2+a^2 \approx a^2$

$F_{Net} = \frac{2kQqy}{(a^2)^{3/2}}$

$\vec{F}_{Net} = +\left(\frac{2kQq}{a^3}\right)\vec{y}$

$x << a$





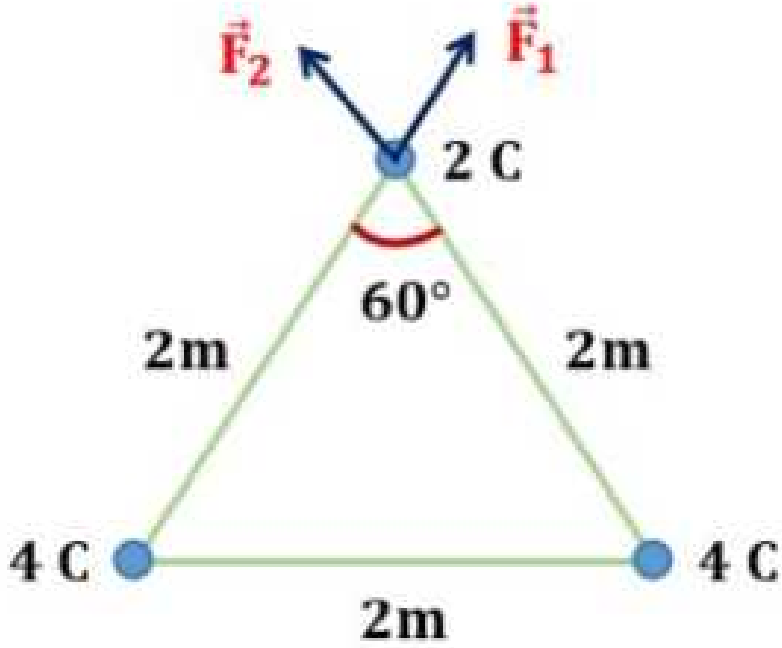
Let's Solve.



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Lets Solve

Q. Find the net force on 2 C charge placed in the figure?
(k is coulomb's constant)



Solution:

F_1 = Force on 2 C due to 4 C on left
F_2 = Force 2 C due to 4 C on right
By principle of superposition;

$$\vec{F}_{net} = \vec{F}_1 + \vec{F}_2$$

$$F_1 = k\frac{(4)(2)}{(2)^2} \quad F_2 = k\frac{(4)(2)}{(2)^2}$$

$$F = (2\sqrt{3}\,k)\text{N}, \quad \text{upwards}$$

Let's Solve.



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Lets Solve

Q. Where should 3rd charge Q be put so that whole system is in equilibrium. Also find nature and value of charge Q.



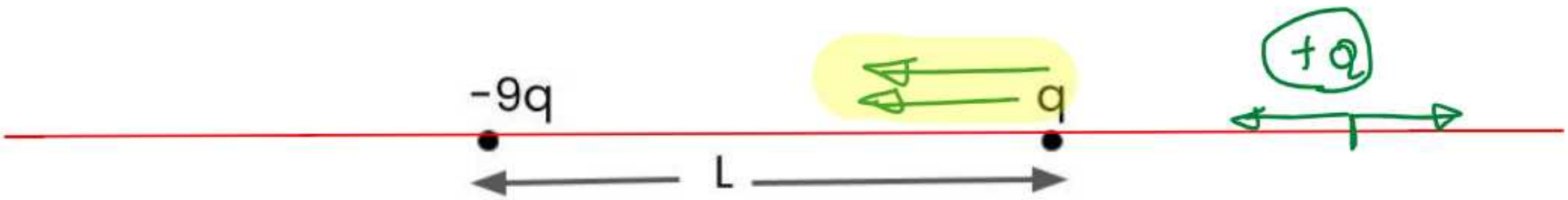
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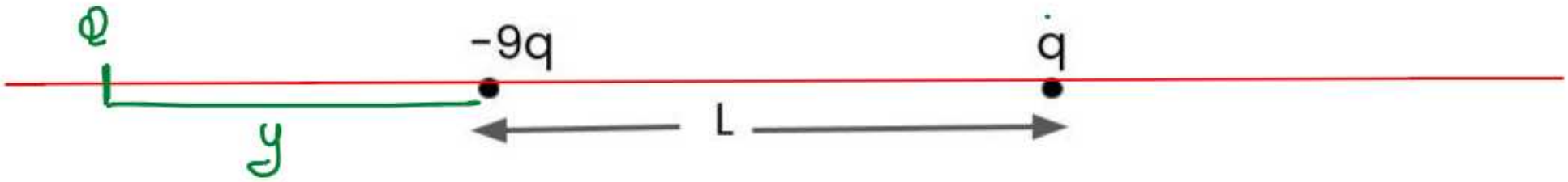
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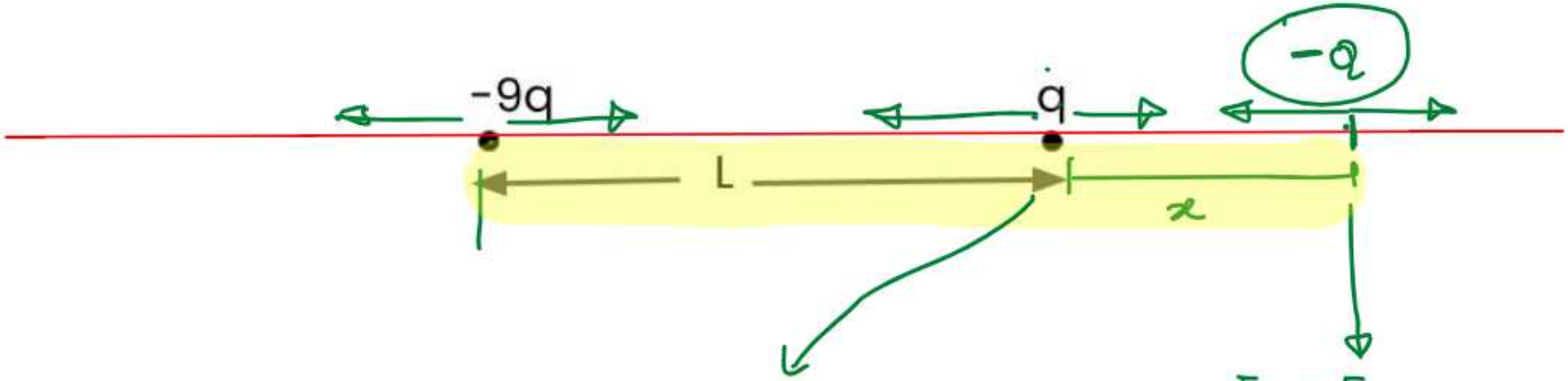
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Lets Solve

Q. Where should 3rd charge Q be put so that whole system is in equilibrium. Also find nature and value of charge Q.



$F_{-9q} = F_{+Q}$

$$\frac{K(9q)(q)}{L^2} = \frac{K(Q)(q)}{\left(\frac{L}{2}\right)^2}$$

$$\frac{9q}{L^2} = \frac{4Q}{L^2}$$

$$Q = \frac{9q}{4}$$

$F_q = F_{-9q}$

$$\frac{Kqq}{x^2} = \frac{K(9q)q}{(L+x)^2}$$

$$\frac{1}{x} = \frac{3}{L+x}$$

$$L + x = 3x$$

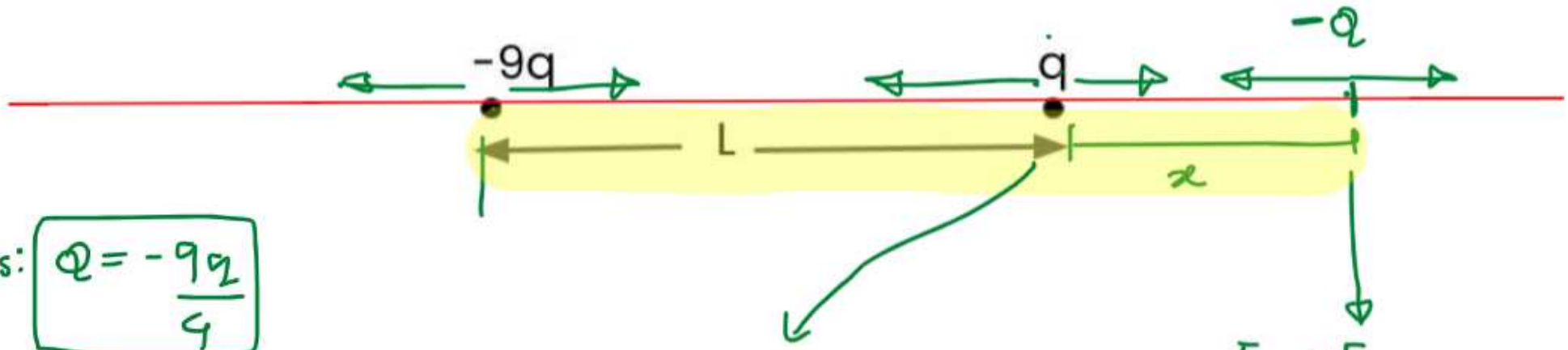
$$L = 3x - x = 2x$$

$$x = \frac{L}{2}$$

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Lets Solve

Q. Where should 3rd charge Q be put so that whole system is in equilibrium. Also find nature and value of charge Q.



Ans: $Q = -\frac{9q}{4}$

$F_{-9q} = F_{+Q}$

$\frac{k(9q)(q)}{L^2} = \frac{k(Q)(q)}{\left(\frac{L}{2}\right)^2}$

$\frac{9q}{L^2} = \frac{4Q}{L^2}$

$Q = \frac{9q}{4}$

$F_q = F_{-9q}$

$\frac{kQq}{x^2} = \frac{k(9q)(q)}{(L+x)^2}$

$\frac{1}{x} = \frac{3}{L+x}$

$L + x = 3x$

$L = 3x - x = 2x$

$x = \frac{L}{2}$

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Lets Solve

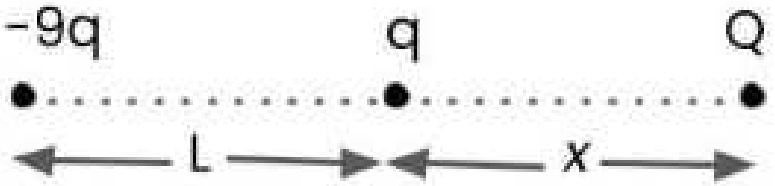
Q. Where should 3rd charge Q be put so that whole system is in equilibrium. Also find nature and value of charge Q.

Soln:

Third charge to be in equilibrium should be placed at sides of charge of less magnitude.

For Q to be in equilibrium,

$$\frac{k9q\cdot Q}{(L+x)^2} \longleftarrow Q \longrightarrow \frac{kqQ}{x^2}$$



$$\frac{k9Qq}{(L+x)^2} = \frac{kqQ}{x^2}$$

$$9x^2 = (L+x)^2$$

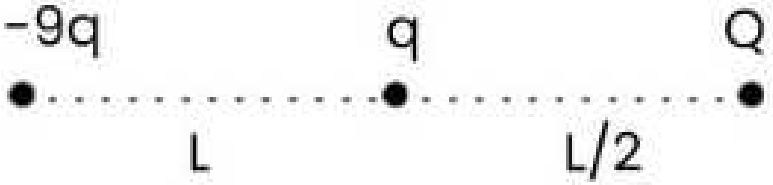
$$x = \frac{L}{2}$$

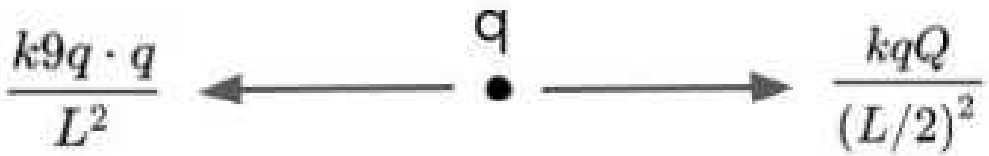
For q to be in equilibrium Q should be -ve in nature.

Lets Solve

Q. Where should 3rd charge Q be put so that whole system is in equilibrium. Also find nature and value of charge Q.

Soln:

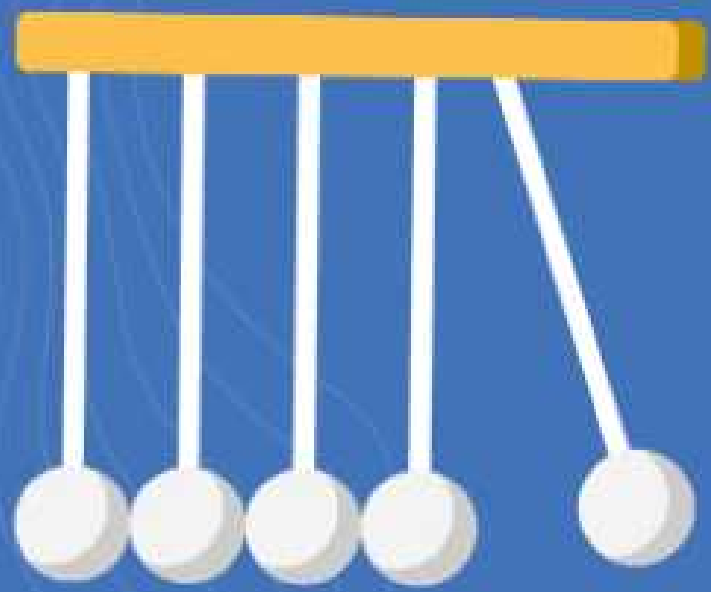




$$\therefore \frac{Kq^2 \cdot 9}{L^2} = \frac{4kqQ}{L^2}$$

$$Q = \frac{9}{4}q \qquad \boxed{\therefore \; Q = -\frac{9}{4}q}$$

Quiz.



Quiz

Chat On

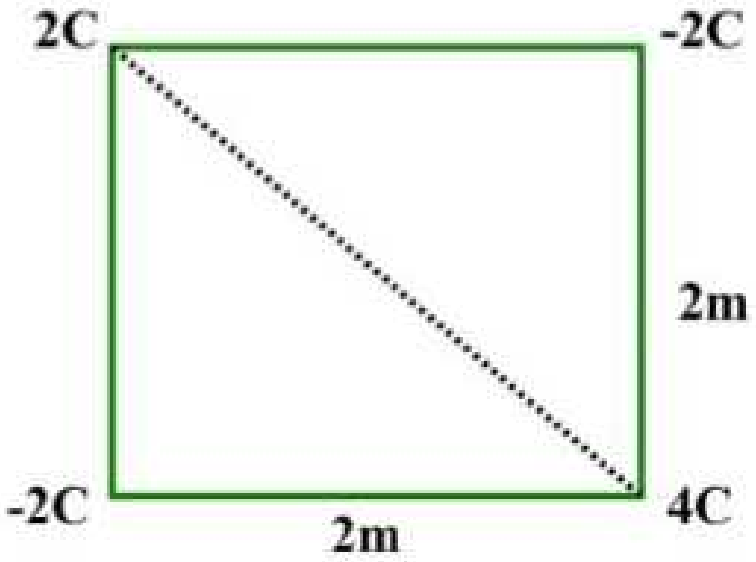
Ex. Find the electrostatic force on 2C, if the point charges are arranged on the vertices of a square ?

A $K\left(\sqrt{2}-1\right)$

B $K\left(\sqrt{2}-2\right)$

C $\sqrt{2}K$

D zero



Quiz

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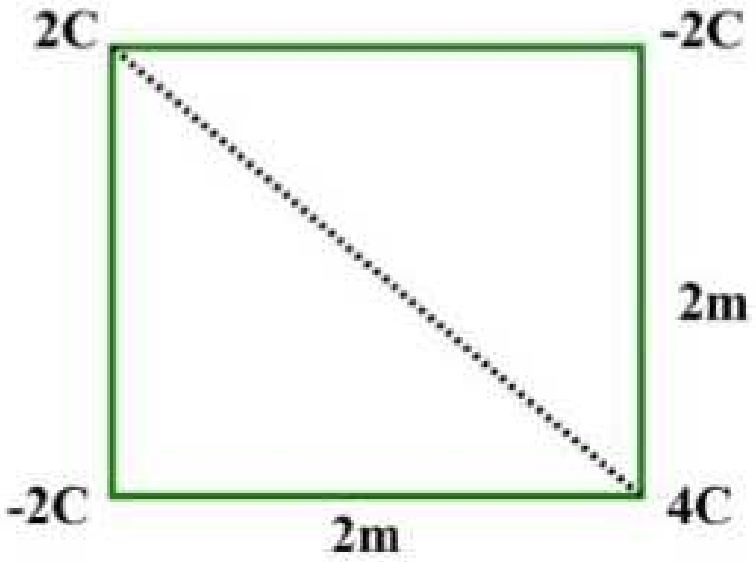
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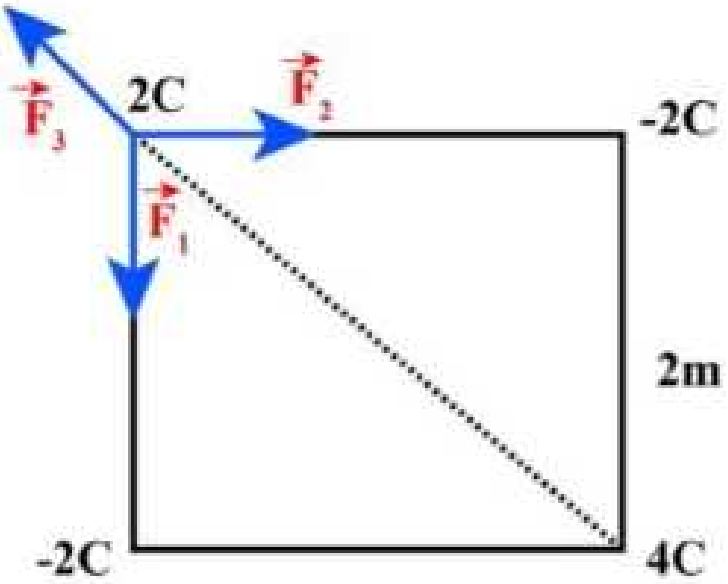
Ex. Ex. Find the electrostatic force on 2C, if the point charges are arranged on the vertices of a square ?

Solution:

$\vec{F}_1$ = Force on 2 C due to -2C below

$\vec{F}_2$ = Force on 2 C due to -2C on right

$\vec{F}_3$ = Force on 2 C due to 4 C on diagonally opposite end by Principle Superposition;



By principle of Superposition;

$$\vec{F}_{net} = \vec{F}_1 + \vec{F}_2 + \vec{F}_3$$

Quiz

Chat On

Ex. Ex. Find the electrostatic force on 2C, if the point charges are arranged on the vertices of a square ?

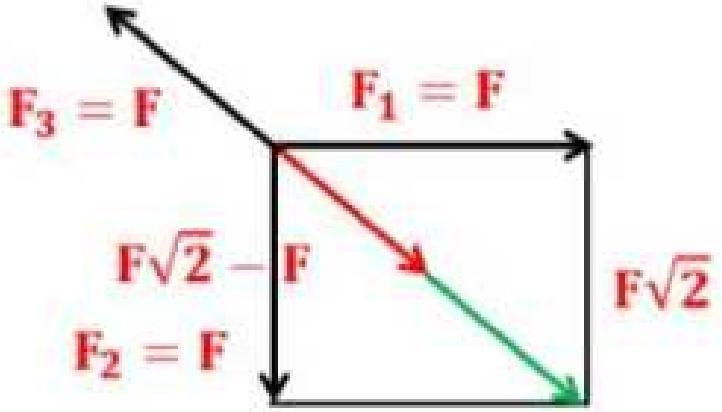
Solution:

$$\vec{F}_1 = K\frac{(2)(2)}{(2)^2} \quad \downarrow$$

$$\vec{F}_2 = K\frac{(2)(2)}{(2)^2} \quad \rightarrow$$

$$\vec{F}_3 = K\frac{(4)(2)}{(2\sqrt{2})^2} \quad \nwarrow$$

$$F_{net} = K\left(\sqrt{2}-1\right)$$

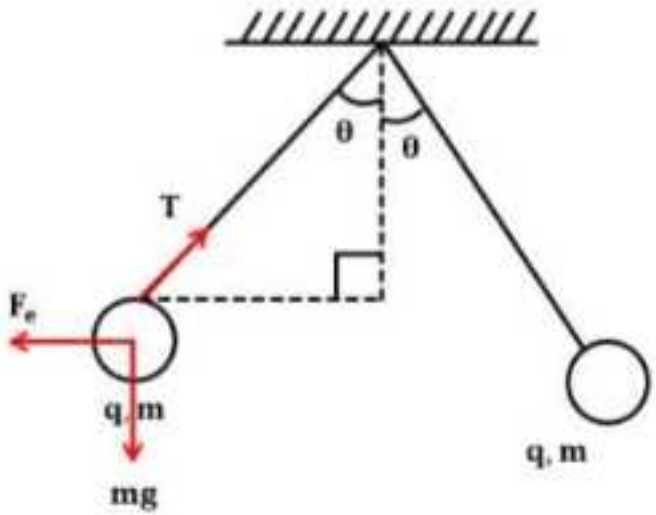


$$F_{net} = F\left(\sqrt{2}-1\right)$$

Quiz

Chat Off

Ex: Two identical small charged spheres, each having a mass "m", hang in equilibrium as shown. The length of each string is "l", and the angle made by any string with the vertical is θ. Find the magnitude of the charge on each sphere?



A $q = (4\ell \sin\theta)\sqrt{\pi\varepsilon_0 mg \tan\theta}$

B $q = (\ell \sin\theta)\sqrt{\pi\varepsilon_0 mg \tan\theta}$

C $q = (\ell \cos\theta)\sqrt{\pi\varepsilon_0 mg \cos\theta}$

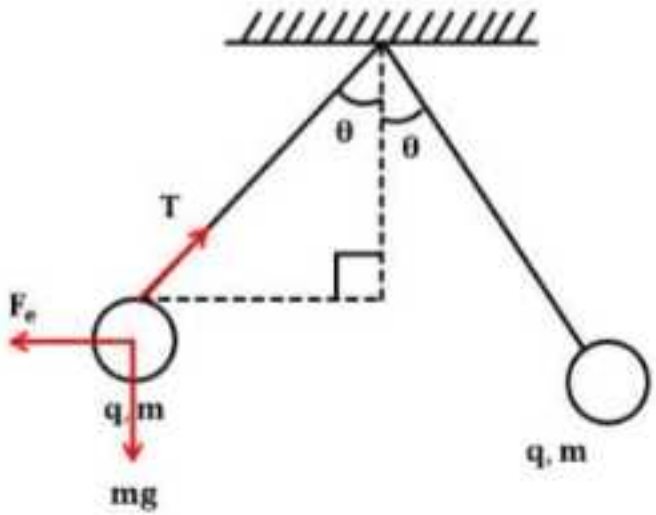
D zero

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Quiz

Chat On

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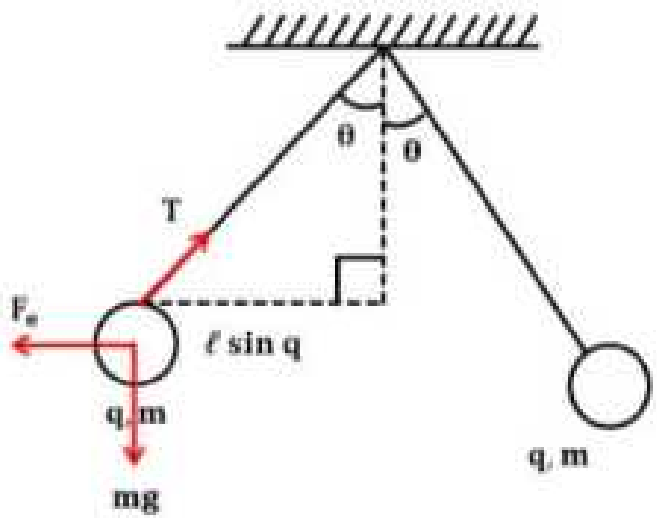
D zero

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Quiz

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Quiz

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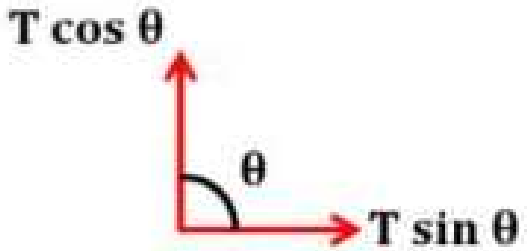
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Solution:

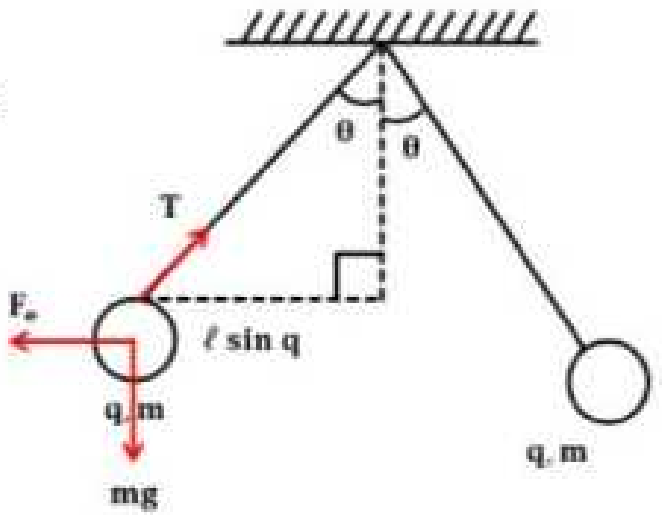
Equilibrium implies net force on the boc

Equate all forces on any body to zero

Step1: Draw FBD of any body



$$F_e = \frac{kq^2}{(2/\sin q)^2}$$



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Quiz

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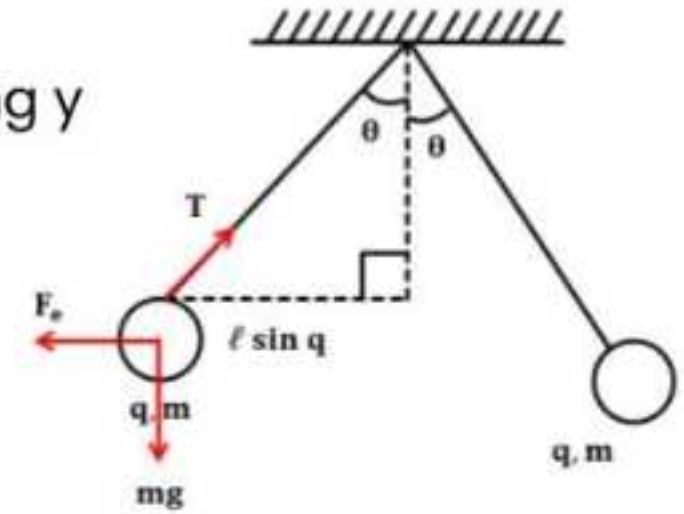
Ex: Two identical small charged spheres, each having a mass "m", hang in equilibrium as shown. The length of each string is "l", and the angle made by any string with the vertical is θ. Find the magnitude of the charge on each sphere?

Solution:

Step 2: Writing forces along x and along y

$$F_x = T\sin\theta - \frac{kq^2}{(2/\sin q)^2}$$

$$F_y = T\cos\theta - mg$$



Ex: Two identical small charged spheres, each having a mass "m", hang in equilibrium as shown. The length of each string is "l", and the angle made by any string with the vertical is θ. Find the magnitude of the charge on each sphere?

Solution:

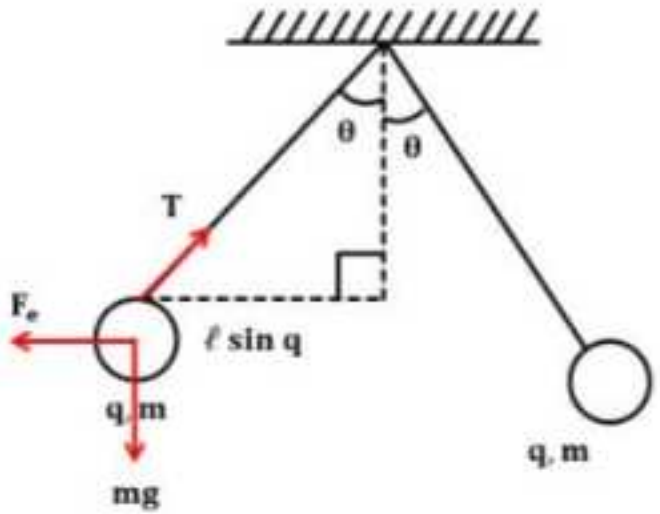
Step 3: Applying equilibrium condition

$$F_x = 0;\ F_y = 0$$

$$T\sin\theta - \frac{kq^2}{(2l\sin\theta)^2} = 0;\ T\cos\theta - mg = 0$$

$$\Rightarrow T\sin\theta = \frac{kq^2}{(2l\sin\theta)^2} \quad \ldots.(1)$$

$$T\cos\theta = mg$$

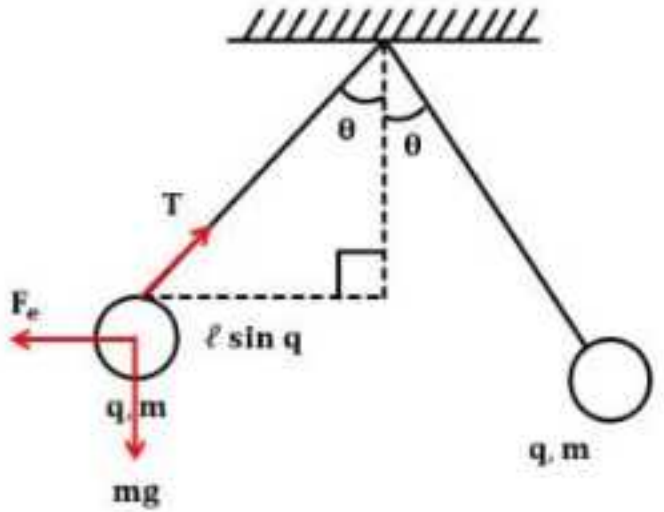


Quiz

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Ex: Two identical small charged spheres, each having a mass "m", hang in equilibrium as shown. The length of each string is "l", and the angle made by any string with the vertical is θ. Find the magnitude of the charge on each sphere?

Solution:



$$\tan\theta = \frac{kq^2}{(mg)\,(2l\sin\theta)^2}$$

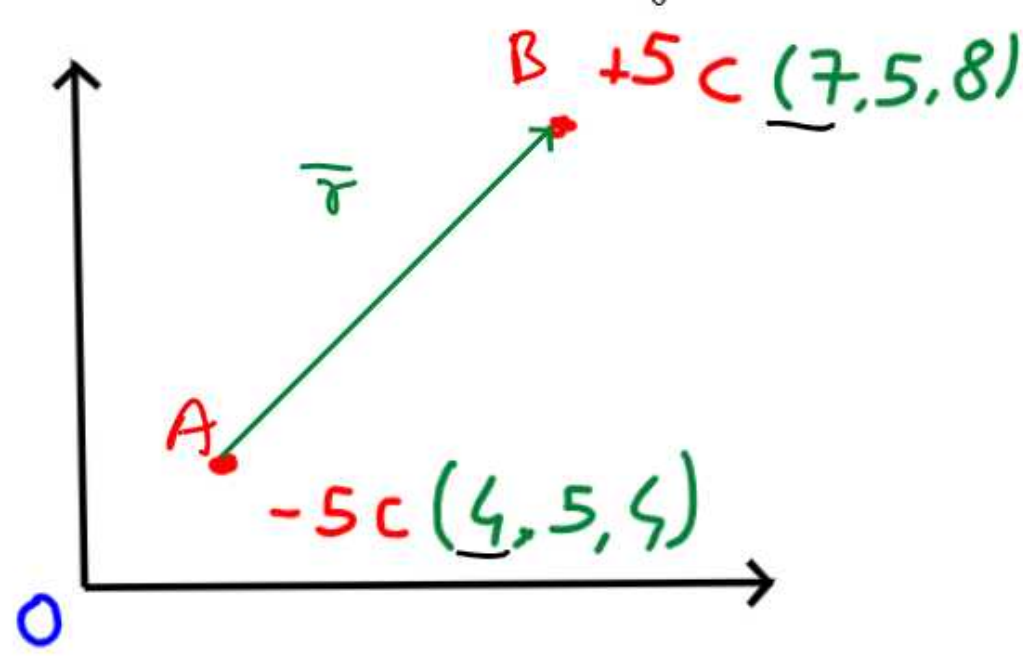
$$\Rightarrow \frac{kq^2}{4l^2 mg\sin^2\theta} = \tan\theta$$

$$\therefore\ q = \sqrt{\frac{4l^2 mg\sin^2\theta\tan\theta}{k}} = \sqrt{16\pi\varepsilon_0 l^2 mg\sin^2\theta\tan\theta}$$

$$or\ q = (4l\sin\theta)\sqrt{\pi\varepsilon_0 mg\tan\theta}$$

Vedantu


r
q1
q2



$$\bar{F} = \frac{kq_1q_2\,\bar{r}}{r^3}$$

$$\bar{F} = \frac{kq_1q_2}{r^2}\hat{r}$$

$$= \frac{kq_1q_2}{r^2}\left[\frac{\bar{r}}{r}\right]$$

Q. Find force acting on +5C due to -5C.

$$\bar{r} = \overline{AB} = \bar{B} - \bar{A}$$

$$\bar{r} = (7-4)\hat{i} + (5-5)\hat{j} + (8-4)\hat{k}$$

$$\bar{r} = 3\hat{i} + 0\hat{j} + 4\hat{k}$$

$$r = |\bar{r}| = \sqrt{3^2+4^2} = 5$$

$$\bar{F} = \frac{k(5)(-5)(3\hat{i}+4\hat{k})}{5^3}$$

$$\bar{F} = -\frac{9\times10^9}{5}(3\hat{i}+4\hat{k})$$

Quiz

Chat Off

Ex. Two charges $\sqrt{3} \times 10 - 19C$ and -10-6 C are placed at (0, 0, 0) and (1, 1, 1) m respectively. Find the force on second charge in vector form.

A $3 \times 10^{-16} \left(\hat{i} - \hat{j} + \hat{k}\right) N$

B $3 \times 10^{-16} \left(\hat{i} + \hat{j} + \hat{k}\right) N$

C $-3 \times 10^{-16} \left(\hat{i} + \hat{j} + \hat{k}\right) N$

D $3 \times 10^{-16} \left(\hat{i} - \hat{j} - \hat{k}\right) N$

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Quiz

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Ex. Two charges $\sqrt{3} \times 10 - 19C$ and -10-6 C are placed at (0, 0, 0) and (1, 1, 1) m respectively. Find the force on second charge in vector form.

Solution:

$$\vec{F}_{21} = \frac{1}{4\pi\epsilon_0}\frac{q_1 q_2}{r_{12}^2}\hat{r}_{12}$$

$$\vec{r}_{12} = (1-0)\hat{i} + (1-0)\hat{j} + (1-0)\hat{k} = \hat{i} + \hat{j} + \hat{k}$$

$$|\vec{r}_{12}| = \sqrt{1^2 + 1^2 + 1^2} = \sqrt{3}$$

$$\hat{r}_{12} = \frac{\vec{r}_{12}}{|\vec{r}_{12}|} = \frac{(\hat{i} + \hat{j} + \hat{k})}{\sqrt{3}}$$

$$\vec{F}_{21} = \frac{9 \times 10^9 \times \sqrt{3} \times 10^{-19} \times (-10^{-6})}{3}\frac{\hat{i} + \hat{j} + \hat{k}}{\sqrt{3}}$$

$$= -3 \times 10^{-16}(\hat{i} + \hat{j} + \hat{k})N$$

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Quiz

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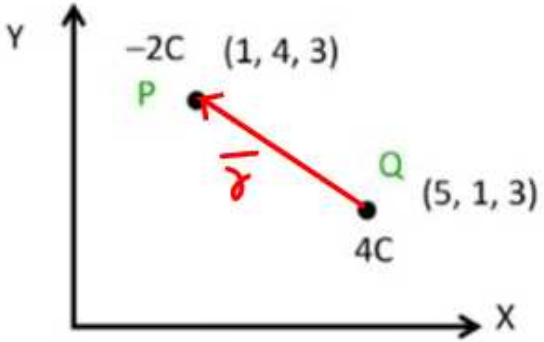
Ex. Find electric force on -2 C due to 4 C ?

A $5.76 \times 10^8 \left(4\hat{i} - 3\hat{j}\right) N$

B $5.76 \times 10^8 \left(4\hat{i} + 3\hat{j}\right) N$

C $5.76 \times 10^8 \left(-4\hat{i} + 3\hat{j}\right) N$

D $5.76 \times 10^8 \left(-4\hat{i} - 3\hat{j}\right) N$



$\vec{r} = \vec{QP} = \vec{P} - \vec{Q}$

$= (1-5)\hat{i} + (4-1)\hat{j} + (0)\hat{k}$

$\vec{r} = -4\hat{i} + 3\hat{j}$

$|\vec{r}| = \sqrt{4^2 + 3^2} = 5$

$\vec{F} = \frac{k q_1 q_2 \vec{r}}{r^3}$

$= \frac{9 \times 10^9 \times -2 \times 4}{125} (-4\hat{i} + 3\hat{j})$

$= \frac{72 \times 10^9}{125} (4\hat{i} - 3\hat{j})$

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Quiz

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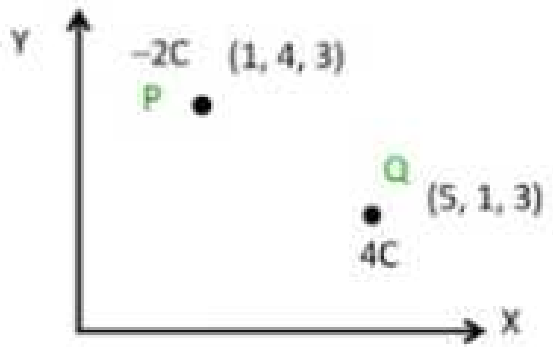
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C $5.76 \times 10^8 \left(-4\hat{i} + 3\hat{j}\right) N$

D $5.76 \times 10^8 \left(-4\hat{i} - 3\hat{j}\right) N$



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Quiz

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Ex. Find electric force on -2 C due to 4 C ?

Solution:

$$\vec{F}_{12} = \frac{kq_1q_2\left(\vec{r}_1 - \vec{r}_2\right)}{\left|\vec{r}_1 - \vec{r}_2\right|^3}$$

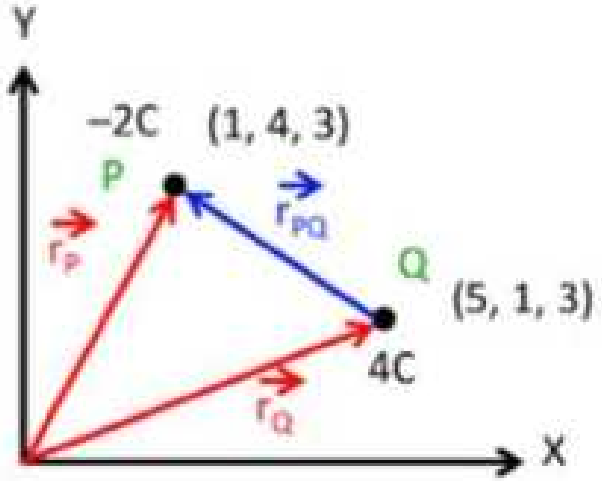
Our Objective is to find $\vec{F}_{PQ}$

$$\vec{r}_P = 1\hat{i} + 4\hat{j} + 3\hat{k}$$

$$\vec{r}_Q = 5\hat{i} + \hat{j} + 3\hat{k}$$

$$\vec{r}_{PQ} = \vec{r}_P - \vec{r}_Q = -4\hat{i} + 3\hat{j} + 0\hat{k}$$

$$\vec{F}_{PQ} = \frac{kq_1q_2\,\vec{r}_{PQ}}{\left|\vec{r}_{PQ}\right|^3}$$



$$\Rightarrow \vec{F}_{PQ} = \frac{9\times10^9\times(-2)(4)\left(-4\hat{i}+3\hat{j}\right)}{\left(\sqrt{(4)^2+(3)^2}\right)^3}$$

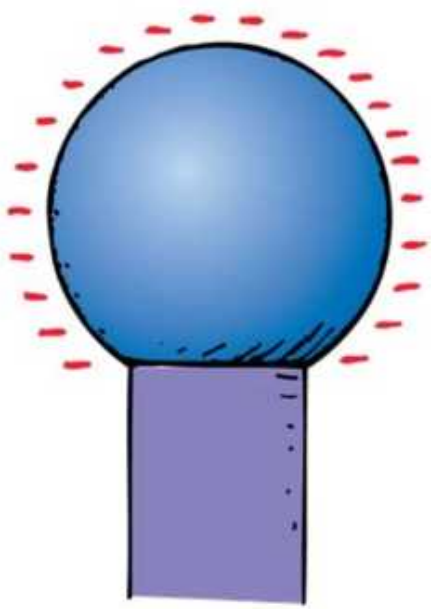
$$\Rightarrow \vec{F}_{PQ} = 5.76\times10^8\left(4\hat{i} - 3\hat{j}\right)N$$

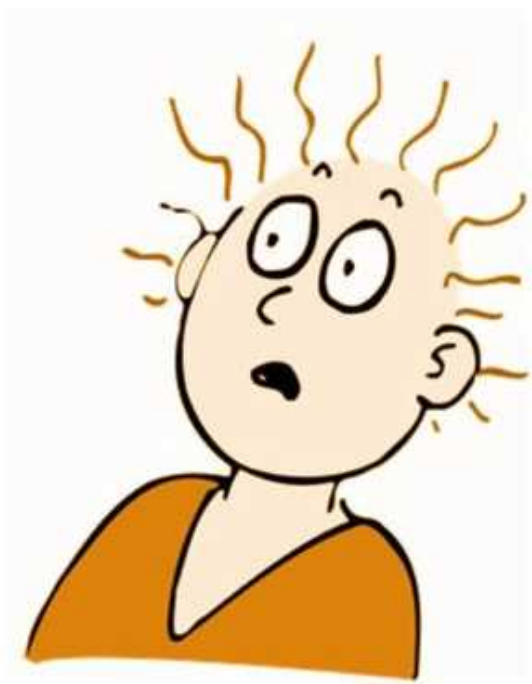
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Electric field

If you walk by the charged dome of an electrostatic machine (for example a Van de Graaff generator), you can sense the charge. Hair on your body stands out—just a tiny bit if you're more than a meter away, and more if you're closer.

The space is said to contain an electric force field (or just, electric field).





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PHYSICS-EL-02 → Q1 to Q15

Summary.



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Summary.

- Force due to multiple charges and Vector form of Coulomb's law and Role of dielectric.
- Introduction to Electric field and Electric field due to point charge
- Electric field due to a system of charges and superposition principle

Electric field

① Due to a point charge



$$E \text{ at } p = \frac{kQ}{r^2}$$

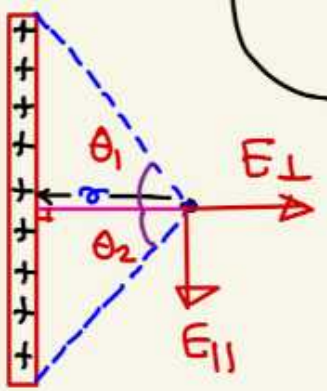
② Due to rod

→ (I) Point on axis



$$E \text{ at } p = \frac{kQ}{r(L+r)}$$

→ (II) Point Not on axis



$$E_{11} = \frac{k\lambda}{r}[\cos\theta_2 - \cos\theta_1]$$

$$E_{\perp} = \frac{k\lambda}{r}[\sin\theta_2 + \sin\theta_1]$$

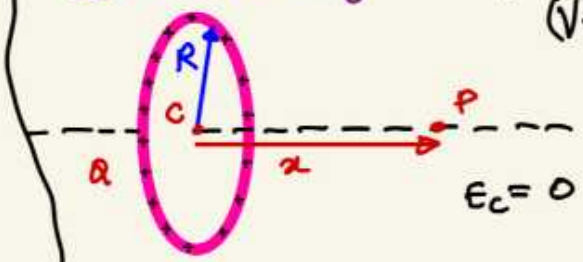
③ Due to arc



$$E = \frac{2k\lambda}{R}\sin\left(\frac{\phi}{2}\right)$$

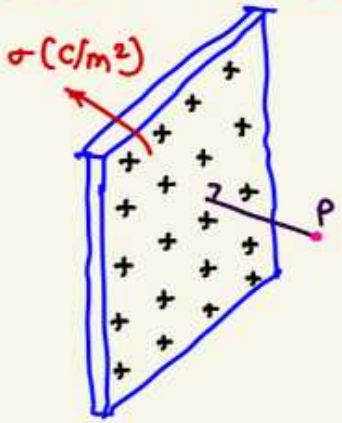
④ Due to Ring

$$E_p = \frac{kQx}{(\sqrt{x^2+R^2})^3}$$



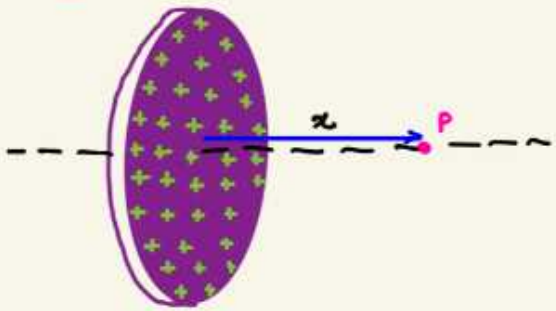
$$E_C = 0$$

⑤ Due infinite sheet



$$E_p = \frac{\sigma}{2\varepsilon_0}$$

⑥ Disc



$$E_p = \frac{\sigma}{2\varepsilon_0}\left[1 - \frac{x}{\sqrt{R^2+x^2}}\right]$$

→ Infinitly long rod, $E_{11} = 0$, $E_{\perp} = \frac{2k\lambda}{r}$] → $E_{Net} = \frac{2k\lambda}{r}$

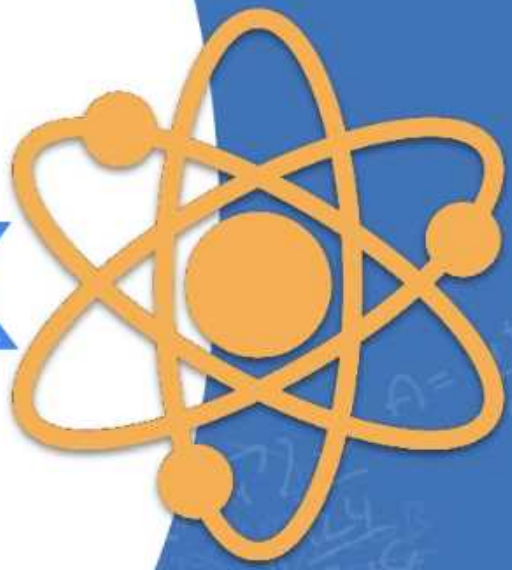
→ Semi Infinite rod $E_{11} = E_{\perp} = \frac{k\lambda}{r}$] → $E_{Net} = \frac{\sqrt{2}k\lambda}{r}$



Thank You.

Share your feedback it matters.

Feedback



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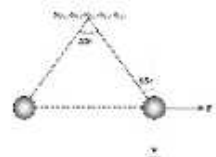
PHYSICS-EL-02

Topics Covered: *Coulombs Law and Idea of Electric Field*

Total Count of Questions: *21*

Expected Time of Attempt by Students: *105 min*

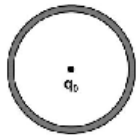
1. Charges 5.0×10^{-7}C, 2.5×10^{-7}C and 1.0×10^{-7}C are fixed at the three corners A, B and C of an equilateral triangle of side 5.0 cm. Find the electric force on the charge at C due to the rest two.
2. A Ten charged particles are kept fixed on the X-axis at point x = 10 mm, 20 mm, 30 mm,.....100 mm. The first particle has a charge 10^{-8}C, the second 8×10^{-8}C, the third 27×10^{-8}C and so on. The tenth particle has a charge 1000×10^{-8}C. Find the magnitude of electric force acting on a 1 C charge placed at the origin.
3. Two identically charged spheres are suspended by strings of equal length. The strings make an angle of 30° with each other. When suspended in a liquid of density 0.8 gm/cc the angle remains same. What is the dielectric constant of liquid.
   Density of sphere = 1.6 gm/cc.



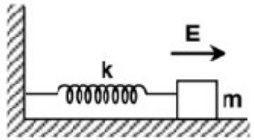
4. Given a cube with point charges q on each of its vertices. Calculate the magnitude of force exerted on any of the charges due to rest of the 7 charges.
5. Two particles, each having a mass of 5 gm and charge 1.0×10^{-7} C, stay in limiting equilibrium on a horizontal table with a separation of 10 cm between them. The coefficient of friction between each particle and the table is the same. Find the value of this coefficient. (Neglect gravitational force between the particles)
6. Two particles A and B having charges q and 2q respectively are placed on a smooth table with a separation d. A third particle C is to be clamped on the table in such a way that the particles A and B remain at rest on the table under electrical forces. What should be the charge on C and where should it be clamped?
7. Two identical pith balls are charged by rubbing against each other. They are suspended from a horizontal rod through two strings of length 20 cm each. The separation between the suspension points being 5 cm. In equilibrium the separation between the balls is 3 cm. Find the mass m of each ball and the tension in the strings. The charge on each ball has magnitude 2×10^{-8}C.
8. A charge Q is to be divided on two small objects. What should be the value of the charges on the objects so that the force between the objects will be maximum.
9. Two free charge +q and +4q are placed apart at a distance ℓ. Find the magnitude, sign and location of a third which makes the system in equilibrium.
10. Two negative charges of unit magnitude and a positive charge q are placed along straight line. At what position and for what value of q will the system be in equilibrium?
11. Two identical positive point charge Q each are fixed apart at a distance 2a. A point charge q lies midway between the fixed charges. Show that
    (a) For small displacement (relative to a) along line joining the fixed charges, the charge q executes SHM if it is +ve and find the time period of oscillations.
    (b) For small lateral displacement, it executes SHM if it is –ve. And find the time period of oscillations.
12. Two equal positive point charges are separated by a distance 2a. A point test charge is located in a plane which is normal to the line joining these charges and midway between them.
    (a) Calculate the radius r of the circle of symmetry in this plane for which the force on the test charge has a maximum value.
    (b) What is the direction of this force, assuming a positive test charge?
13. A charge Q is placed at each of two opposite corners of a square. A charge –q is placed at each of the other two corners. If the resultant force on Q is zero, how are Q and q related.
14. Three balls, each of mass 10 gm are suspended separately from a point by silk threads each one meter long. The balls are identically charged and hang at the corners of an equilateral triangle of side 0.1 metre. What is the charge on each ball?

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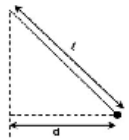
15. A ring of radius R with a uniformly distributed charge q is shown in figure. A charge q_0 is now placed at the centre of the ring. Find the increment in the tension in ring.



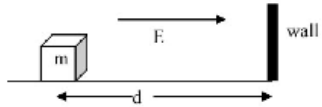
16. A block of mass m and having a charge q is placed on a smooth horizontal table and is connected to a wall through an unstressed spring of spring constant k as shown in figure. A horizontal electric field E parallel to the spring is switched on.
Find the amplitude of the resulting SHM of the block.



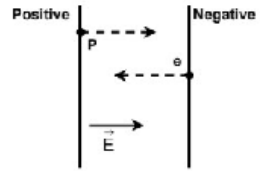
17. A pith ball having a mass m hangs by a fine silk thread ℓ metre long in a horizontal electric field E. When the ball is given an electric charge q coulomb, it stands out d metre from the vertical line. Show that the electric field is given by $E = \frac{mgd}{q\sqrt{\ell^2 - d^2}}$.



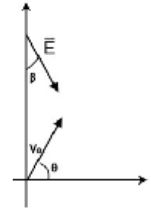
18. In the figure shown, the block has a mass m and charge q. The horizontal surface is frictionless. At $t = 0$, uniform electric field E is applied in the direction shown. If the collision of block with wall is perfectly elastic then find the time-period of motion of block. Is the motion of block S.H.M.?



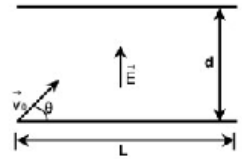
19. Two large parallel copper plates are 'd' distance apart and have a uniform electric field between them as depicted in figure. An electron(mass m_e) is released from the negative plate at the same time that a proton(mass m_p) is released from the positive plate. Neglect the force of the particles on each other and find their distance from the positive plate when they pass each other. (Does it surprise you that you need not know the electric field to solve this problem)?



20. A projectile of mass m and having charge q is thrown with initial velocity v_0 at an angle α with the horizontal. There is an electric field of strength E at angle β with the downward vertical away from the point of projection. Find the time of flight and range of projectile.



21. In figure, a uniform, upward-directed electric field $\vec{E}$ of magnitude 2.00×10^3 N/C has been set up between two horizontal plates by charging the lower plate positively and the upper plate negatively. The plates have length L = 10.0 cm and separation d = 2.00 cm. An electron is then shot between the plates from the left edge of the lower plate. The initial velocity $\vec{v}_0$ of the electron makes an angle $\theta = 45°$ with the lower plate and has a magnitude of 6.00×10^4 m/s.
(a) Will the electron strike one of the plates?
(b) If so, which plate and how far horizontally from the left edge will the electron strike?



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ANSWERS

1. 0.238
2. $4.95 \times 10^{7} N$
3. 2
4. $\frac{kq^2}{a^2}\left[\left(\frac{1}{3\sqrt{3}}+\frac{1}{\sqrt{2}}+1\right)\sqrt{3}\right]$
5. $\mu = 0.18$
6. $\left(\sqrt{2}-1\right)d$ from A towards B and $Q_c = -q\left(6-\sqrt{2}\right)$
7. 7.96 gm, $7.72 \times 10^{-2} N$
8. $\frac{Q}{2}, \frac{Q}{2}$
9. $-\frac{4q}{9}$, at $\frac{l}{3}$ distance from +q (towards +4q)
10. ($q = \frac{1}{4}C$, Midway between -ve charges)
11. (a) $2\pi\sqrt{\frac{ma^3}{4Qqk}}$

    (b) $2\pi\sqrt{\frac{ma^3}{2Qqk}}$
12. (a) $\frac{a}{\sqrt{2}}$

    (b) radius outwards from the centre of the circle
13. $q = -2\sqrt{2}q$
14. $6.2 \times 10^{-8} C$
15. $\frac{kqq_0}{2\pi R^2}$
16. $\frac{qE}{k}$
17. ()

18. $2\sqrt{\frac{2md}{qE}}$, No

19. $\frac{m_e d}{m_e + m_p}$ from positive plate

20. $T_f = \frac{2V_0 \sin\alpha}{\frac{qE}{m}\cos\beta + g}, R = V_0 \cos\alpha \left(\frac{2V_0 \sin\alpha}{\frac{qE}{m}\cos\beta + g}\right) + \frac{1}{2}\left(\frac{qE\sin\beta}{m}\right)\left(\frac{2V_0 \sin\alpha}{\frac{qE}{m}\cos\beta + g}\right)^2$

21. (a) Yes

(b) Upper plate 2.73 cm